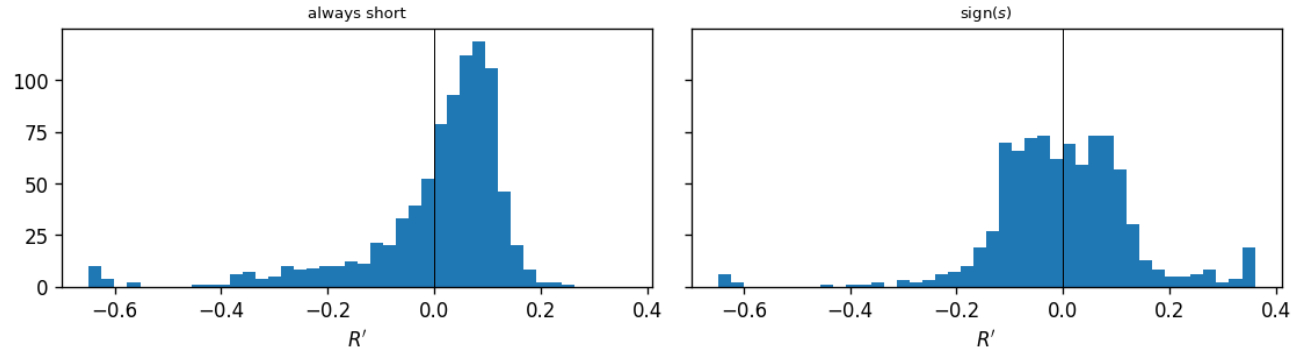


0DTE nearest-OTM package: rule table by strategy, every intraday entry window

Rule table by strategy — entry 10:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{ann}	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>										
all models (no forecast)	864	0.002	0.149	-1.12	0.25	-2.71	10.66	0.49	0.26	-3.74
<i>sign(s) (± 1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	864	0.002	0.149	-1.12	0.78	-0.64	10.51	0.38	0.21	-3.88
block-diagonal ridge	864	0.002	0.149	-1.12	0.78	-0.63	10.51	0.46	0.25	-3.86
block-diagonal ridge, without the FOMC columns	864	0.002	0.149	-1.12	0.78	-0.50	10.50	0.45	0.24	-3.86
LightGBM [†]	864	0.005	0.149	-1.12	0.78	-0.62	10.58	0.98	0.53	-3.61
XGBoost [†]	864	0.003	0.149	-1.12	0.78	-0.62	10.53	0.58	0.32	-3.80
lasso (causally tuned) [†]	864	-0.002	0.149	-1.12	0.78	-0.73	10.44	-0.36	-0.19	-4.25
lasso ($\alpha = 10^{-4}$)	864	0.002	0.149	-1.12	0.78	-0.64	10.51	0.38	0.21	-3.89
elastic net (causally tuned) [†]	864	-0.003	0.149	-1.12	0.78	-0.73	10.42	-0.59	-0.32	-4.36

block-diagonal ridge, midpoint fills: entry 10:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



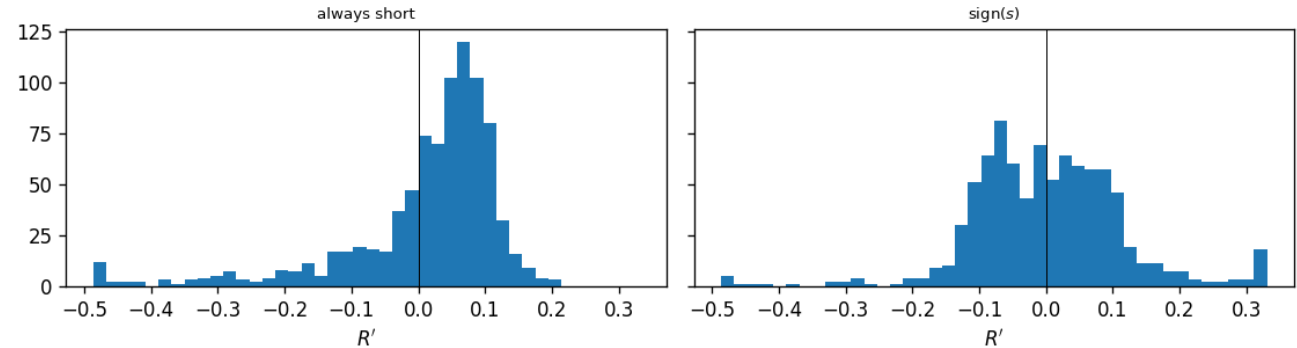
Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 10:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 864 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 10:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	865	0.006	0.154	-1.96	0.21	-5.02	43.98	1.18	0.64	-2.44
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	865	0.002	0.154	-1.96	1.30	-0.77	43.06	0.29	0.16	-2.97
block-diagonal ridge	865	0.003	0.154	-1.96	1.30	-0.76	43.10	0.50	0.27	-2.85
block-diagonal ridge, without the FOMC columns	865	0.002	0.154	-1.96	1.30	-0.74	43.08	0.41	0.22	-2.91
LightGBM [†]	865	0.004	0.154	-1.96	1.30	-0.61	43.16	0.82	0.44	-2.68
XGBoost [†]	865	0.002	0.154	-1.96	1.30	-0.67	43.07	0.34	0.18	-2.94
lasso (causally tuned) [†]	865	-0.003	0.154	-1.96	1.30	-0.73	43.00	-0.56	-0.30	-3.41
lasso ($\alpha = 10^{-4}$)	865	0.004	0.154	-1.96	1.30	-0.77	43.14	0.68	0.37	-2.76
elastic net (causally tuned) [†]	865	-0.002	0.154	-1.96	1.30	-0.76	43.00	-0.43	-0.23	-3.35

block-diagonal ridge, midpoint fills: entry 10:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



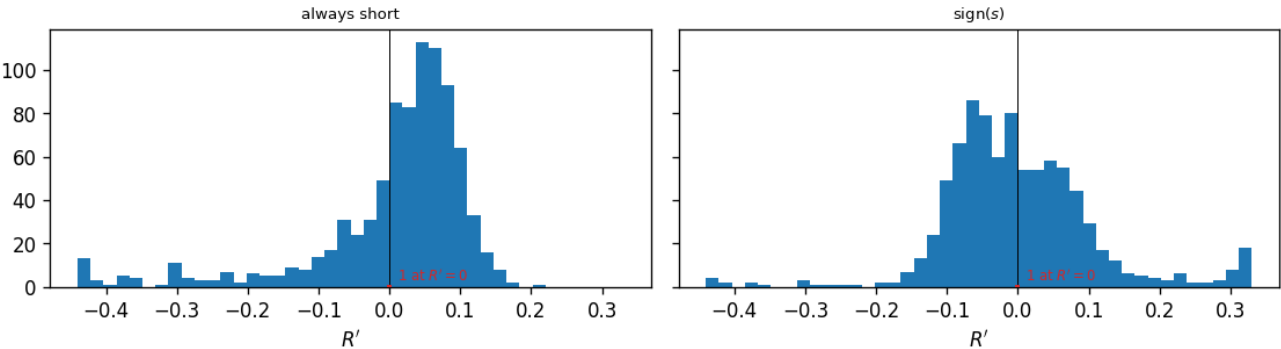
Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 10:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>										
all models (no forecast)	865	0.004	0.131	-1.27	0.21	-3.41	18.52	0.86	0.46	-3.21
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	865	0.004	0.131	-0.79	1.27	1.79	17.91	0.86	0.46	-3.29
block-diagonal ridge	865	0.002	0.131	-0.79	1.27	1.62	17.99	0.50	0.27	-3.49
block-diagonal ridge, without the FOMC columns	865	0.001	0.131	-0.79	1.27	1.64	18.03	0.25	0.13	-3.63
LightGBM [†]	865	0.003	0.131	-0.79	1.27	1.69	17.97	0.59	0.32	-3.45
XGBoost [†]	865	0.003	0.131	-0.79	1.27	1.61	17.97	0.61	0.33	-3.43
lasso (causally tuned) [†]	865	0.003	0.131	-0.79	1.27	1.73	17.94	0.73	0.39	-3.38
lasso ($\alpha = 10^{-4}$)	865	0.002	0.131	-0.79	1.27	1.65	18.00	0.40	0.21	-3.55
elastic net (causally tuned) [†]	865	0.003	0.131	-0.79	1.27	1.65	17.94	0.77	0.41	-3.35

block-diagonal ridge, midpoint fills: entry 11:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



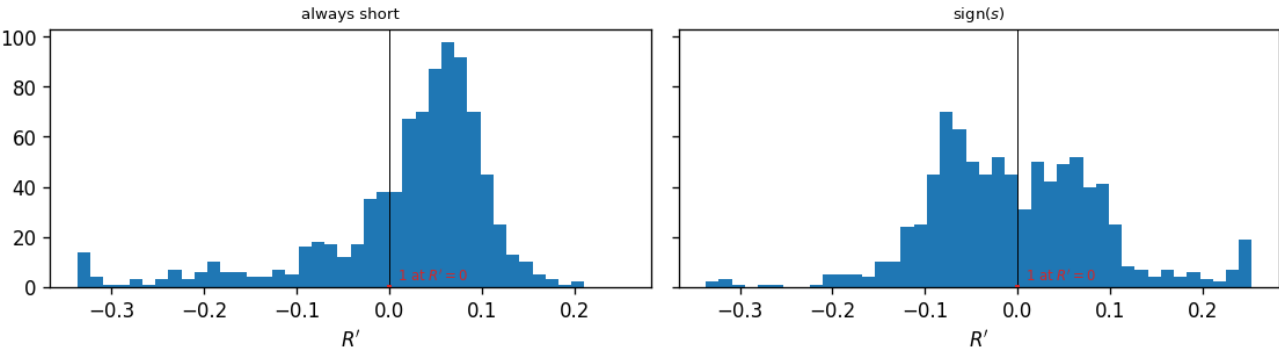
Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 11:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	865	0.016	0.114	-1.06	0.21	-3.37	19.30	4.05	2.19	-2.13
sign(<i>s</i>) (± 1 on the sign of <i>s</i>)										
baseline (HAR + calendar OLS)	865	0.003	0.116	-0.65	1.06	2.04	16.63	0.66	0.36	-4.04
block-diagonal ridge	865	0.001	0.116	-0.65	1.06	2.03	16.73	0.22	0.12	-4.27
block-diagonal ridge, without the FOMC columns	865	-0.002	0.116	-0.65	1.06	2.11	16.95	-0.50	-0.27	-4.66
LightGBM [†]	865	-0.000	0.116	-0.65	1.06	2.00	16.82	-0.09	-0.05	-4.44
XGBoost [†]	865	-0.000	0.116	-0.65	1.06	2.04	16.82	-0.10	-0.05	-4.44
lasso (causally tuned) [†]	865	0.000	0.116	-0.65	1.06	2.15	16.76	0.12	0.06	-4.33
lasso ($\alpha = 10^{-4}$)	865	-0.000	0.116	-0.65	1.06	2.05	16.80	-0.05	-0.03	-4.42
elastic net (causally tuned) [†]	865	0.001	0.116	-0.65	1.06	2.09	16.71	0.31	0.17	-4.23

block-diagonal ridge, midpoint fills: entry 11:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



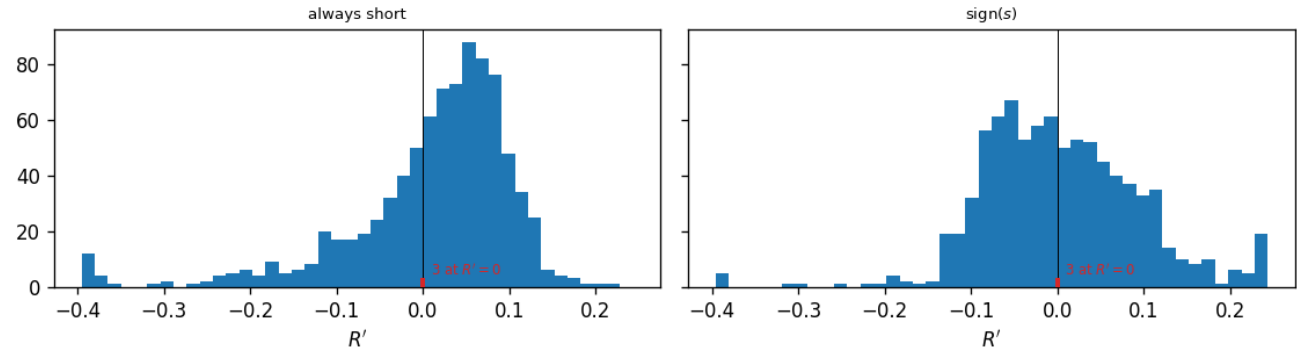
Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 11:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.008	0.112	-1.02	0.22	-3.16	17.88	2.09	1.12	-3.42
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.000	0.112	-1.02	0.54	-1.19	16.81	0.03	0.01	-4.53
block-diagonal ridge	866	0.006	0.112	-1.02	0.84	-0.46	16.98	1.45	0.78	-3.81
block-diagonal ridge, without the FOMC columns	866	0.006	0.112	-1.02	0.84	0.29	16.84	1.52	0.82	-3.78
LightGBM [†]	866	0.001	0.112	-1.02	0.54	-1.27	16.84	0.19	0.10	-4.43
XGBoost [†]	866	-0.000	0.112	-1.02	0.54	-1.25	16.79	-0.09	-0.05	-4.60
lasso (causally tuned) [†]	866	0.005	0.112	-1.02	0.84	0.27	16.84	1.44	0.77	-3.82
lasso ($\alpha = 10^{-4}$)	866	0.005	0.112	-1.02	0.84	-0.46	16.97	1.38	0.74	-3.85
elastic net (causally tuned) [†]	866	0.005	0.112	-1.02	0.84	0.23	16.83	1.30	0.70	-3.91

block-diagonal ridge, midpoint fills: entry 12:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



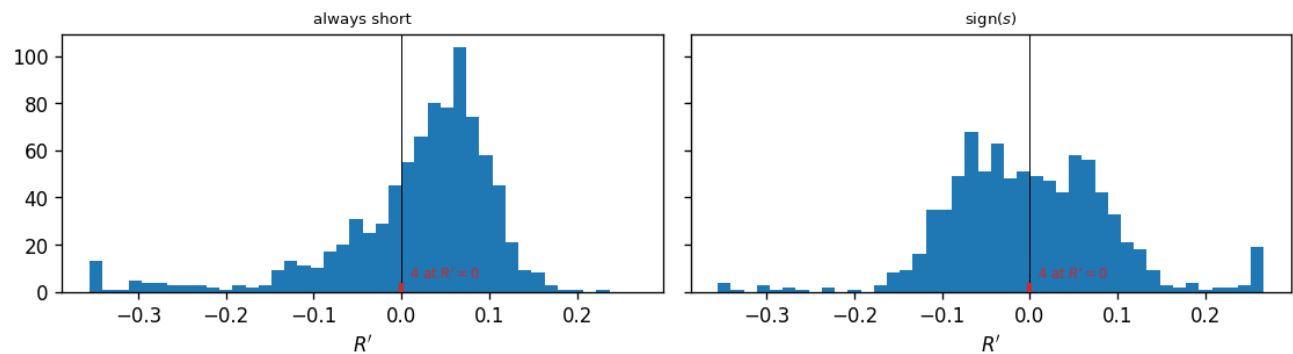
Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 12:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.013	0.115	-1.23	0.23	-3.74	25.06	3.29	1.78	-2.88
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	-0.002	0.116	-1.23	0.78	-0.31	22.80	-0.41	-0.22	-4.86
block-diagonal ridge	866	-0.000	0.116	-1.23	0.78	-0.88	22.79	-0.11	-0.06	-4.70
block-diagonal ridge, without the FOMC columns	866	0.001	0.116	-1.23	0.78	-0.82	22.82	0.15	0.08	-4.56
LightGBM [†]	866	0.001	0.116	-1.23	0.78	-0.22	22.82	0.25	0.13	-4.51
XGBoost [†]	866	-0.000	0.116	-1.23	0.78	-0.26	22.80	-0.07	-0.04	-4.68
lasso (causally tuned) [†]	866	-0.003	0.116	-1.23	0.78	-0.78	22.76	-0.65	-0.35	-4.99
lasso ($\alpha = 10^{-4}$)	866	-0.001	0.116	-1.23	0.78	-0.87	22.78	-0.27	-0.14	-4.78
elastic net (causally tuned) [†]	866	-0.001	0.116	-1.23	0.78	-0.75	22.78	-0.30	-0.16	-4.79

block-diagonal ridge, midpoint fills: entry 12:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



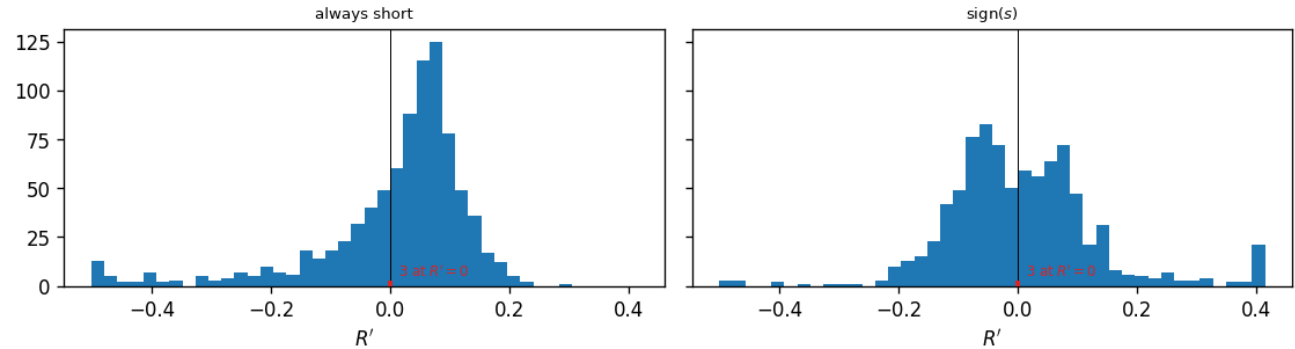
Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 12:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.002	0.168	-1.72	0.29	-4.23	29.08	0.43	0.23	-3.12
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.010	0.168	-0.50	1.72	3.82	28.09	1.82	0.98	-2.43
block-diagonal ridge	866	0.010	0.168	-0.90	1.72	3.31	28.22	1.81	0.98	-2.43
block-diagonal ridge, without the FOMC columns	866	0.013	0.168	-0.51	1.72	3.74	28.00	2.29	1.23	-2.18
LightGBM [†]	866	0.010	0.168	-0.51	1.72	3.79	28.13	1.70	0.92	-2.50
XGBoost [†]	866	0.012	0.168	-0.51	1.72	3.81	28.02	2.10	1.13	-2.29
lasso (causally tuned) [†]	866	0.011	0.168	-0.51	1.72	3.70	28.07	2.01	1.08	-2.33
lasso ($\alpha = 10^{-4}$)	866	0.014	0.168	-0.51	1.72	3.73	27.98	2.42	1.30	-2.11
elastic net (causally tuned) [†]	866	0.010	0.168	-0.51	1.72	3.71	28.12	1.82	0.98	-2.43

block-diagonal ridge, midpoint fills: entry 13:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



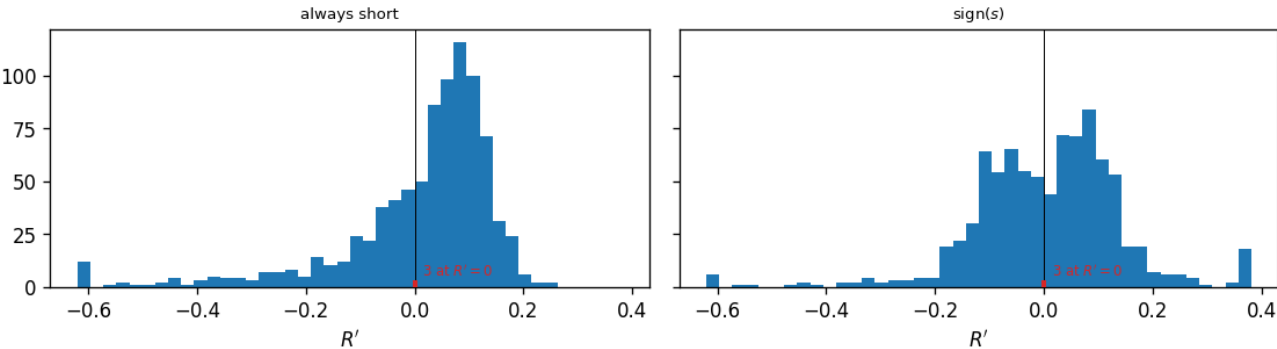
Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 13:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.008	0.161	-1.07	0.25	-2.85	11.71	1.46	0.79	-3.47
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.004	0.161	-1.01	1.07	0.21	11.08	0.70	0.38	-4.00
block-diagonal ridge	866	0.008	0.161	-1.01	1.07	-0.25	11.19	1.47	0.80	-3.58
block-diagonal ridge, without the FOMC columns	866	0.009	0.161	-1.01	1.07	0.14	11.12	1.56	0.84	-3.54
LightGBM [†]	866	0.001	0.161	-1.07	0.96	-0.34	11.10	0.24	0.13	-4.23
XGBoost [†]	866	-0.003	0.161	-1.07	0.96	-0.34	11.07	-0.50	-0.27	-4.62
lasso (causally tuned) [†]	866	0.005	0.161	-1.01	1.07	0.50	11.05	0.96	0.52	-3.86
lasso ($\alpha = 10^{-4}$)	866	0.009	0.161	-1.01	1.07	-0.18	11.21	1.74	0.94	-3.44
elastic net (causally tuned) [†]	866	0.004	0.161	-1.07	0.96	-0.15	11.12	0.74	0.40	-3.96

block-diagonal ridge, midpoint fills: entry 13:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



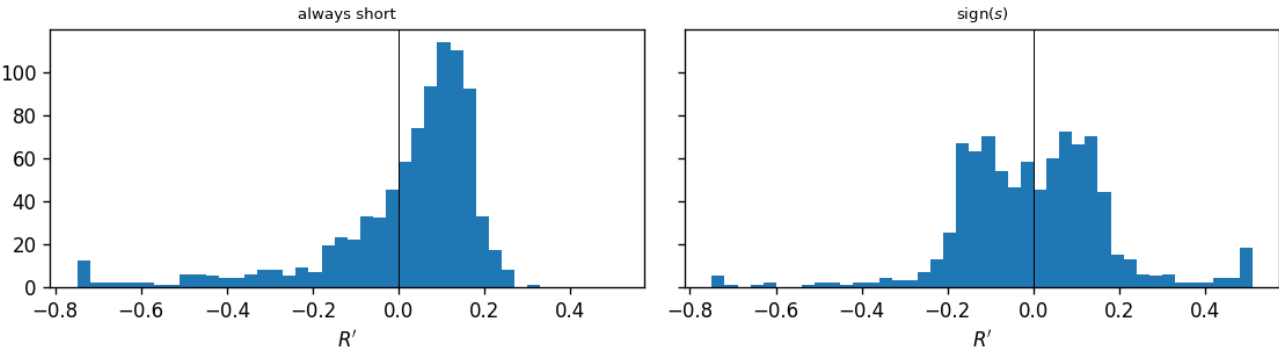
Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 13:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} [×]
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.008	0.237	-2.65	0.32	-4.48	32.87	0.93	0.50	-2.62
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.002	0.237	-2.65	1.63	-1.64	32.29	0.20	0.11	-3.06
block-diagonal ridge	866	0.002	0.237	-2.65	1.63	-1.55	32.31	0.31	0.17	-3.00
block-diagonal ridge, without the FOMC columns	866	0.000	0.237	-2.65	1.63	-1.52	32.24	0.02	0.01	-3.16
LightGBM [†]	866	-0.000	0.237	-2.65	1.63	-1.47	32.23	-0.02	-0.01	-3.19
XGBoost [†]	866	0.001	0.237	-2.65	1.63	-1.48	32.26	0.09	0.05	-3.13
lasso (causally tuned) [†]	866	-0.001	0.237	-2.65	1.63	-1.51	32.23	-0.06	-0.03	-3.20
lasso ($\alpha = 10^{-4}$)	866	0.002	0.237	-2.65	1.63	-1.54	32.29	0.21	0.12	-3.05
elastic net (causally tuned) [†]	866	-0.001	0.237	-2.65	1.63	-1.51	32.21	-0.12	-0.07	-3.24

block-diagonal ridge, midpoint fills: entry 14:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



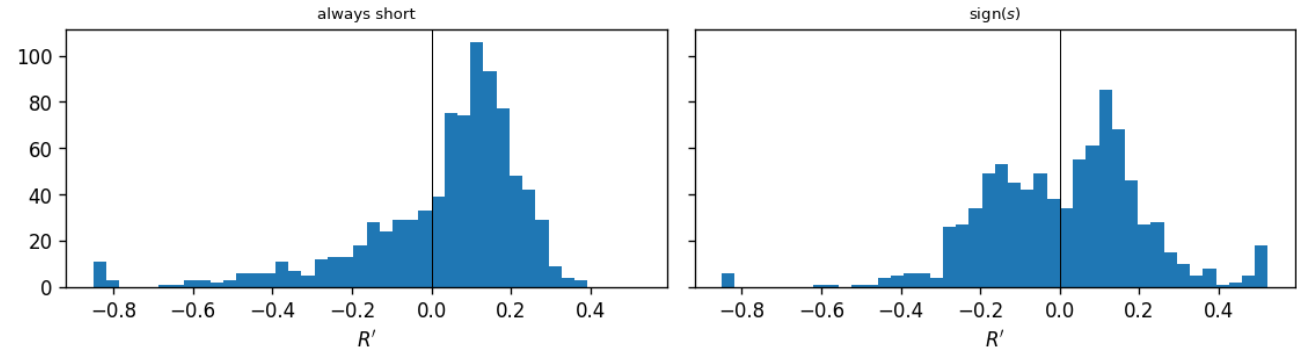
Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 14:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:30 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe ^x _{ann}
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.019	0.300	-5.15	0.38	-7.56	108.09	1.86	1.00	-1.52
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.008	0.300	-5.15	2.01	-5.13	105.97	0.74	0.40	-2.15
block-diagonal ridge	866	0.009	0.300	-5.15	2.01	-5.43	106.17	0.88	0.48	-2.08
block-diagonal ridge, without the FOMC columns	866	0.010	0.300	-5.15	2.01	-5.11	106.25	0.99	0.53	-2.02
LightGBM [†]	866	0.007	0.300	-5.15	2.01	-5.45	105.97	0.71	0.38	-2.17
XGBoost [†]	866	0.005	0.300	-5.15	2.01	-5.39	105.76	0.51	0.28	-2.28
lasso (causally tuned) [†]	866	0.009	0.300	-5.15	2.01	-5.42	106.13	0.85	0.46	-2.09
lasso ($\alpha = 10^{-4}$)	866	0.007	0.300	-5.15	2.01	-5.41	105.93	0.67	0.36	-2.19
elastic net (causally tuned) [†]	866	0.007	0.300	-5.15	2.01	-5.41	105.94	0.68	0.37	-2.19

block-diagonal ridge, midpoint fills: entry 14:30 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



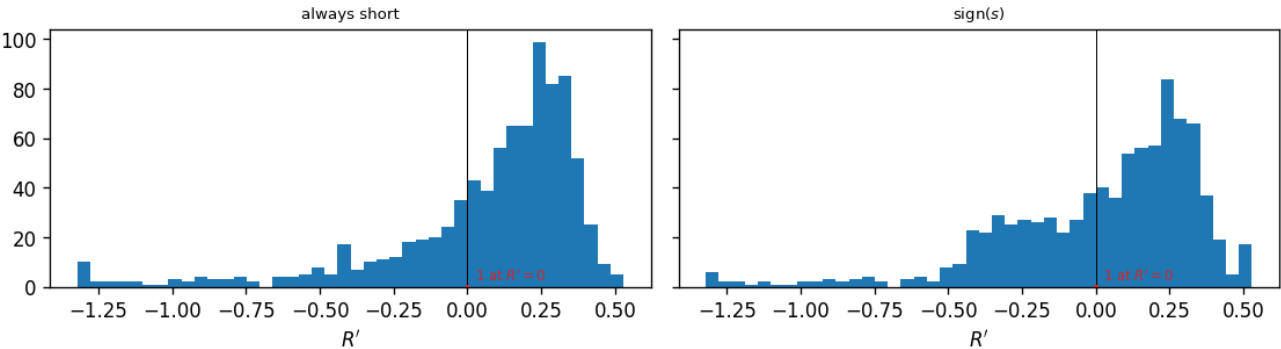
Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 14:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:00 ET, one-bar hold

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.060	0.379	-3.31	0.58	-3.03	15.14	4.64	2.51	0.18
<i>sign(s) (±1 on the sign of s)</i>										
baseline (HAR + calendar OLS)	866	0.021	0.383	-3.31	2.85	-1.13	12.91	1.58	0.85	-1.43
block-diagonal ridge	866	0.030	0.383	-3.31	2.85	-0.94	13.04	2.28	1.23	-1.06
block-diagonal ridge, without the FOMC columns	866	0.031	0.383	-3.31	2.85	-0.94	13.06	2.39	1.29	-1.00
LightGBM [†]	866	0.023	0.383	-3.31	2.85	-1.07	12.95	1.80	0.97	-1.31
XGBoost [†]	866	0.017	0.384	-3.31	2.85	-1.03	12.82	1.27	0.69	-1.59
lasso (causally tuned) [†]	866	0.031	0.383	-3.31	2.85	-0.80	13.02	2.39	1.29	-1.00
lasso ($\alpha = 10^{-4}$)	866	0.030	0.383	-3.31	2.85	-0.94	13.03	2.27	1.22	-1.07
elastic net (causally tuned) [†]	866	0.027	0.383	-3.31	2.85	-1.18	13.06	2.11	1.14	-1.15

block-diagonal ridge, midpoint fills: entry 15:00 ET, one-bar hold; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



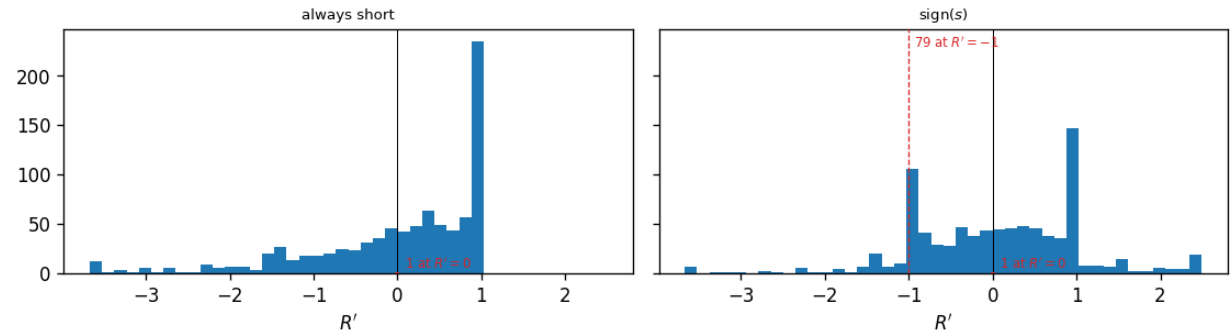
Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM package at 15:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:30 ET, cash-settled at the close

	<i>n</i>	mean	std	min	max	skew	ex. kurt	<i>t</i>	Sharpe _{ann}	Sharpe [×] _{ann}
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.014	1.128	-10.32	1.00	-2.31	10.89	0.38	0.20	-0.26
sign(<i>s</i>) (±1 on the sign of <i>s</i>)										
baseline (HAR + calendar OLS)	866	0.069	1.126	-5.42	10.32	0.58	10.71	1.79	0.97	0.50
block-diagonal ridge	866	0.095	1.124	-5.42	10.32	0.92	10.61	2.48	1.34	0.87
block-diagonal ridge, without the FOMC columns	866	0.090	1.124	-5.42	10.32	1.00	10.59	2.36	1.27	0.80
LightGBM [†]	866	0.103	1.123	-5.42	10.32	1.01	10.58	2.71	1.46	0.99
XGBoost [†]	866	0.106	1.123	-5.42	10.32	0.91	10.62	2.78	1.50	1.03
lasso (causally tuned) [†]	866	0.100	1.123	-6.84	10.32	0.46	10.78	2.62	1.41	0.94
lasso ($\alpha = 10^{-4}$)	866	0.107	1.123	-5.42	10.32	1.22	10.50	2.80	1.51	1.04
elastic net (causally tuned) [†]	866	0.072	1.125	-6.84	10.32	0.46	10.74	1.90	1.02	0.56

block-diagonal ridge, midpoint fills: entry 15:30 ET, cash-settled at the close; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)



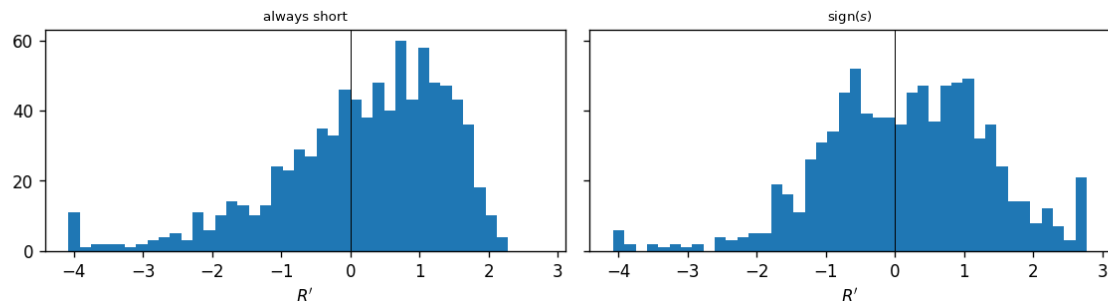
Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the package expires worthless).

One trade a day: enter the nearest-OTM package at 15:30 ET and let it cash-settle at the official close. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck’s frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

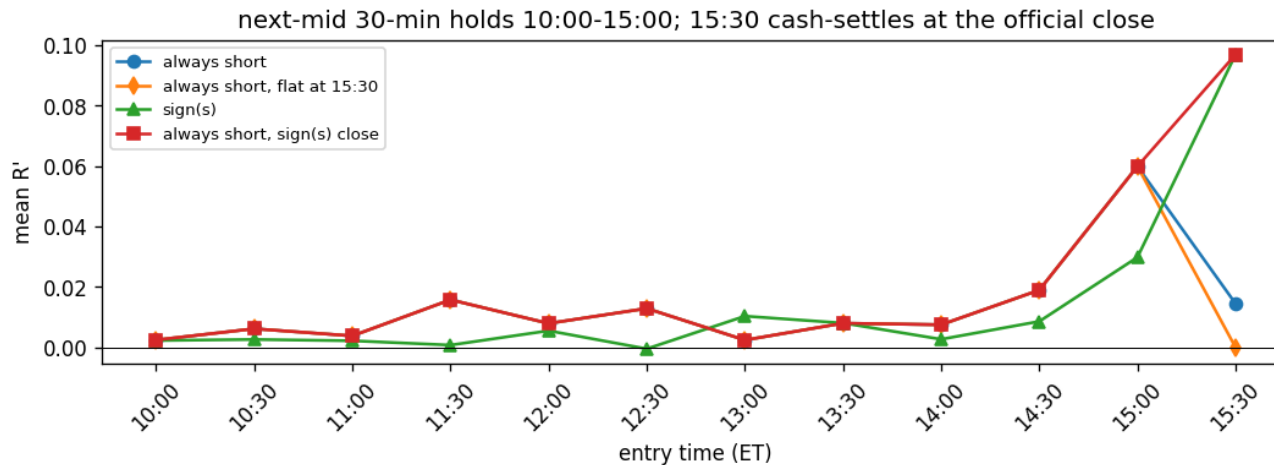
Rule table by strategy — pooled over the twelve entry windows

	n	mean	std	min	max	skew	ex. kurt	t	Sharpe _{ann}	Sharpe _{ann} ^x
<i>Short volatility (always short)</i>										
all models (no forecast)	866	0.160	1.338	-11.40	2.24	-1.92	8.73	3.53	1.90	-1.92
sign(s) (± 1 on the sign of s)										
baseline (HAR + calendar OLS)	866	0.121	1.346	-5.96	11.40	0.27	7.23	2.64	1.43	-2.87
block-diagonal ridge	866	0.167	1.339	-6.26	11.40	0.55	7.53	3.68	1.99	-2.33
block-diagonal ridge, without the FOMC columns	866	0.163	1.334	-6.26	11.40	0.59	7.57	3.59	1.94	-2.39
LightGBM [†]	866	0.158	1.341	-5.96	11.40	0.58	7.62	3.46	1.87	-2.44
XGBoost [†]	866	0.144	1.352	-5.96	11.40	0.47	7.45	3.14	1.69	-2.56
lasso (causally tuned) [†]	866	0.158	1.337	-7.03	11.40	0.30	7.98	3.47	1.87	-2.40
lasso ($\alpha = 10^{-4}$)	866	0.179	1.335	-6.26	10.65	0.60	6.45	3.96	2.13	-2.20
elastic net (causally tuned) [†]	866	0.123	1.360	-7.03	11.40	0.26	7.79	2.67	1.44	-2.78

block-diagonal ridge, midpoint fills: daily sum over the twelve entry windows 10:00-15:30 ET; return of the position, 1st-99th percentile window (bars: the days with $R' \neq 0$)

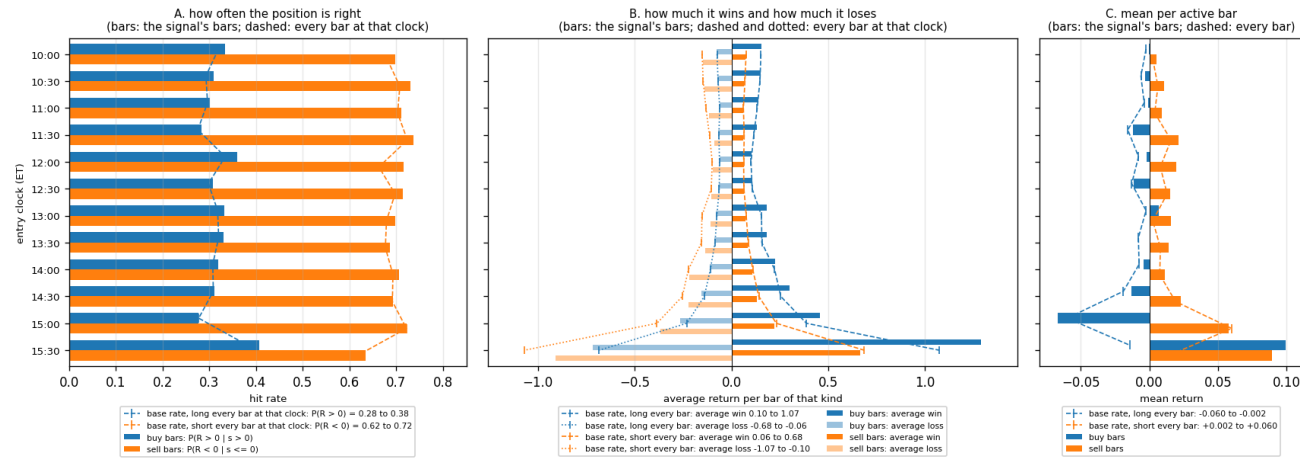


Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' by entry time, the four rules of the intraday notebook: next-mid 30-minute holds 10:00–15:00, the 15:30 leg cash-settling at the official close.

the matched signal's bars against the base rates (the same statistic with the package held every day, no forecast), 866 days, midpoint fills



The matched signal's bars against the base rates — the same statistic with the package held every day, no forecast — by entry time, 866 days, midpoint fills.

Each row is the daily sum over the twelve entry windows 10:00–15:30 ET: one number per expiration day, then the usual statistics on that daily series. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Notes

One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean}/\text{std}$, with no autocorrelation correction. “ex. kurt” is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. $\text{Sharpe}_{\text{ann}} = (\text{mean}/\text{std}) \times \sqrt{252}$; every other column is daily. The one column this table adds to the deck’s is $\text{Sharpe}_{\text{ann}}^{\times}$, the same rule’s annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck’s.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - IV_{\text{hr}}^2 h_t w_t$: the forecast against the implied variance of the same window. On the bars whose vendor implied volatility is a censored solver node the deck’s treatment is used here too — the volatility that reproduces the package midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.